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Protein Synthesis

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What are proteins?

- Proteins are essential quality of macromolecules that contribute to the structure and function of the cell
- Enzymes – that determine metabolic activities
- **One Gene One Enzyme Hypothesis (1941) –** Beadle & Tatum
  - Genes control chemical reactions by regulating or determining the specificities of enzymes
- Before them, Archibald Garrod (1901) – Alkaptonuria – **Inborn Errors of Metabolism –** linked biochemical defects to genes
- PKU – Phenylketonuria (Phenylalanine breakdown missing)

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Messenger to the Gene

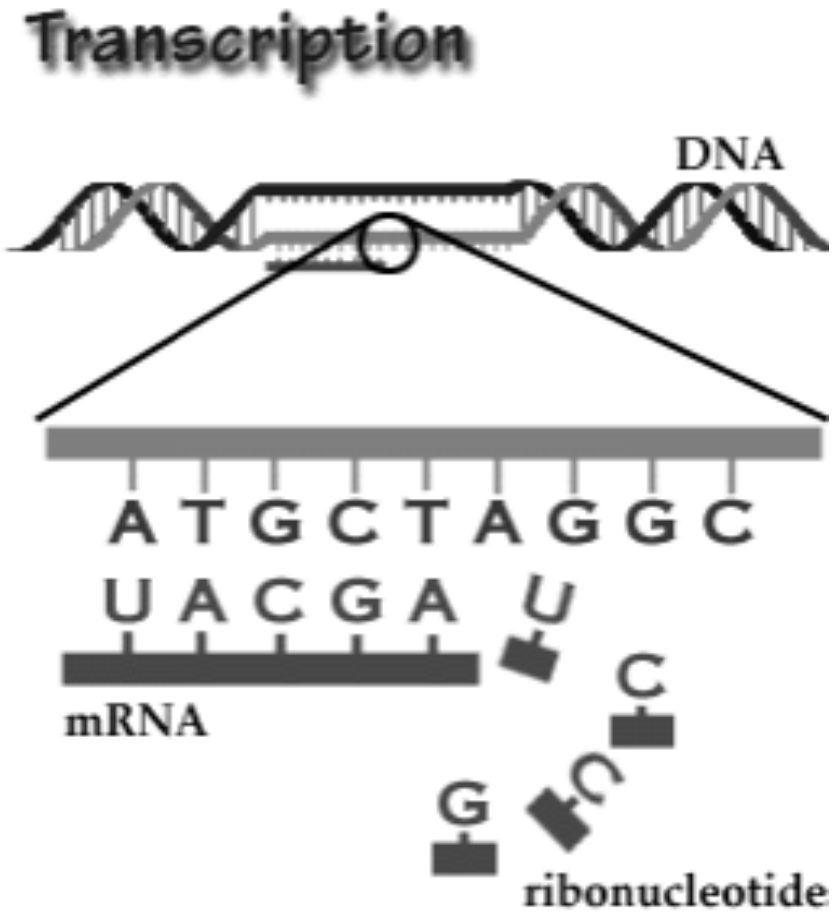
- Protein synthesis occurs in the cytoplasm of the cell
- If genes are responsible for it, then the DNA is located in the nucleus of the cell!
- There must be a molecule that serves as a "messenger" to carry the genetic message from the gene to the ribosome, the organelle associated with protein synthesis in the cytoplasm of cells.
- This molecule is **Messenger RNA or mRNA** which moves from nucleus to cytoplasm and acts as **messenger to the Gene**

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Transcription of mRNA

- mRNA is formed when a gene (particular segment of DNA molecule) becomes active
- In the same manner of DNA replication, an active gene serves as a **template** for the formation of a piece of RNA
- This piece of RNA has sequence of bases **complementary** to one of the strands of DNA and similar to the other strand

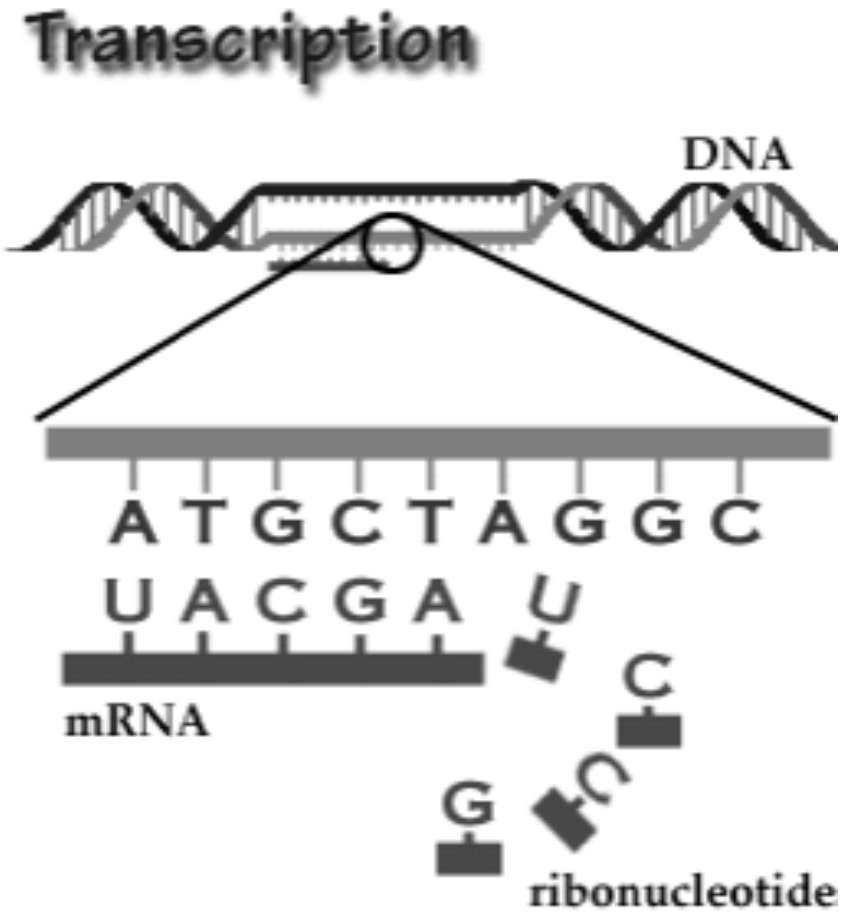


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Transcription

- Each sequence of three bases on the mRNA is called a **codon.**
- The process of copying an RNA strand from a DNA template is known as **transcription.**
- Usually only one strand of the DNA helix serves as s template for transcription.



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Processing of mRNA

- After transcription in the nucleus, certain chemical changes must take place on mRNA before release into the cytoplasm
- One of the changes is called **CAPPING –** involving addition of a Methyl group (CH3) on one end and addition of a number of Adenines (Poly-A) on the other.
- The function of these changes – still not completely known
- May be they help binding of the mRNA with other cytoplasmic factors responsible for Protein Synthesis

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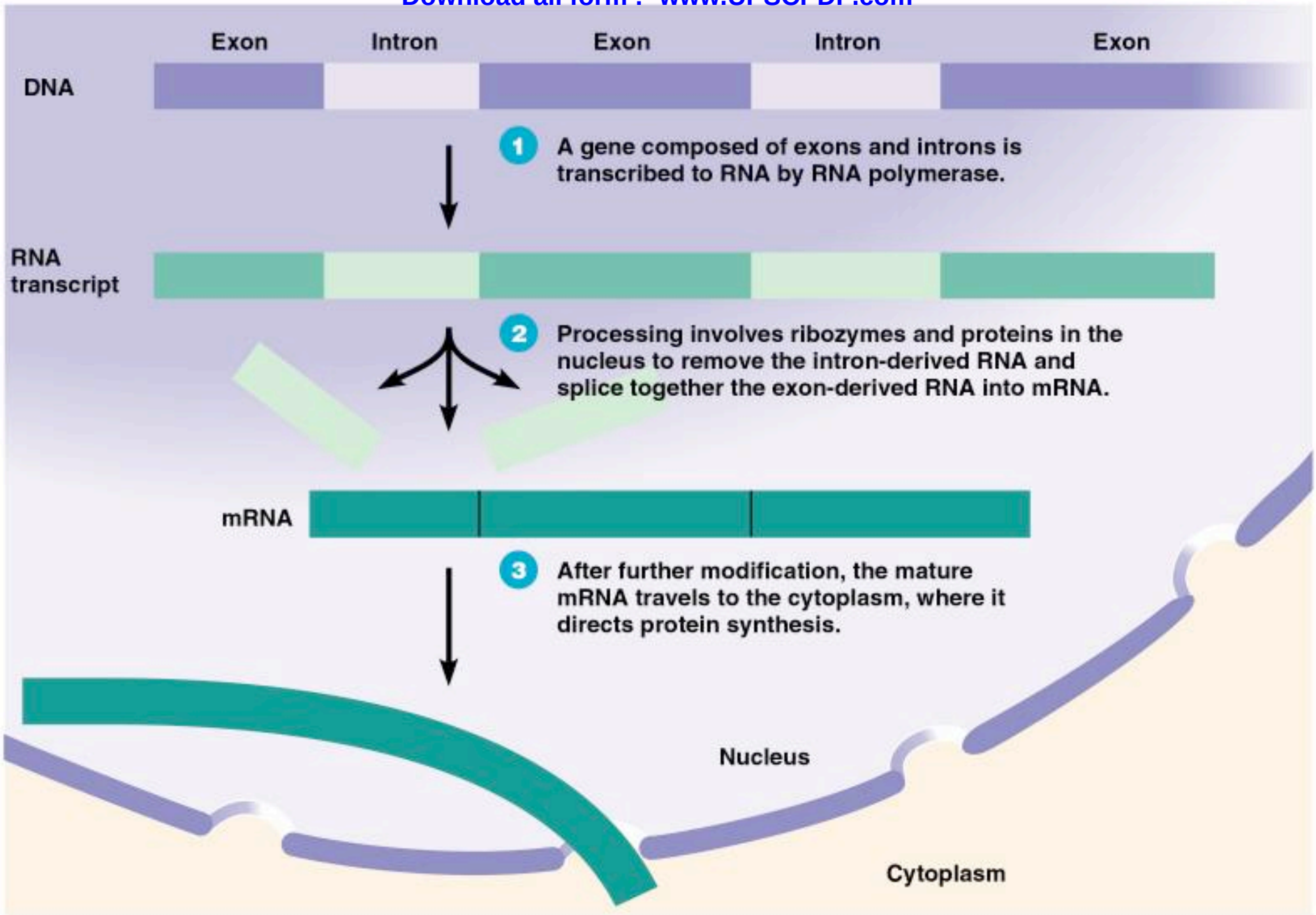
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Processing of mRNA

- It has been discovered that on a newly transcribed mRNA (which is still in the nucleus) the number of bases are more than those on a mature mRNA (which is in the cytoplasm)
- It was later found that portions of the DNA which are transcribed on the mRNA are actually **excised** or removed from mRNA before it leaves the nucleus.
- Ex. mRNA of Alpha 2 Collagen is 50,000 bases long; yet its mature mRNA is only 5000 bases long!
- The area of DNA which is transcribed but is excised is called the **intron** and the sequence retained is **exon**

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Translation of Genetic Material

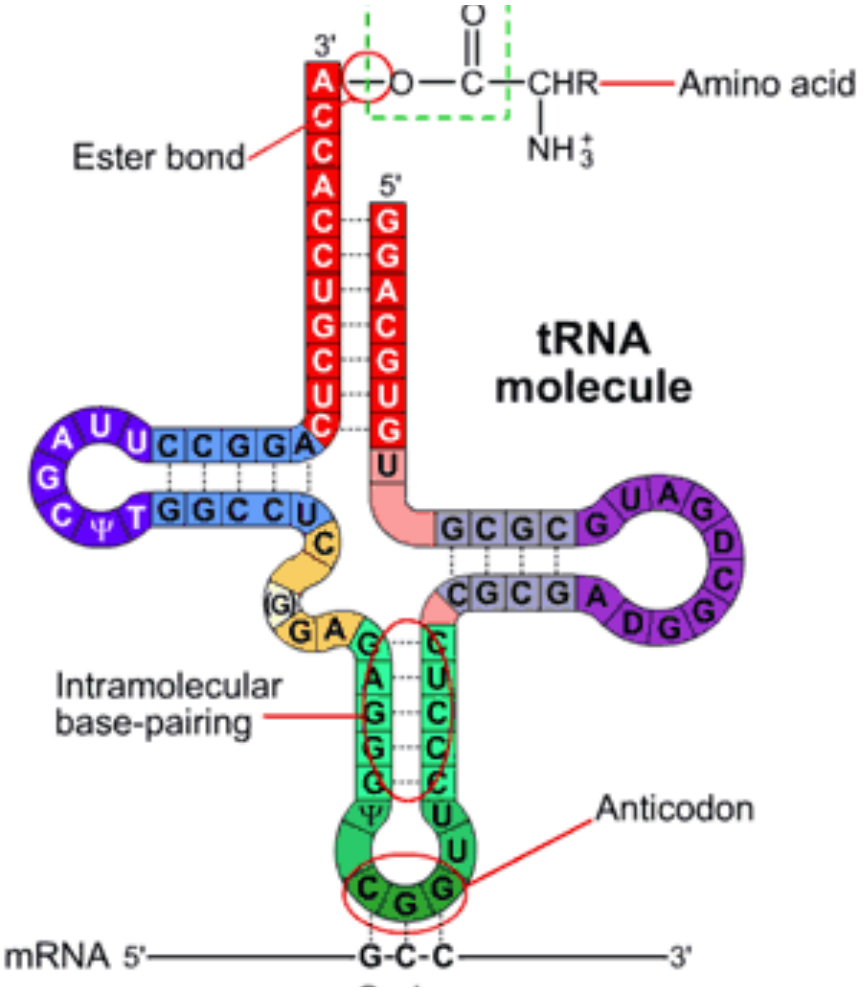
- Once the mRNA leaves nucleus and enters cytoplasm, it binds to the ribosomes
- Ribosomes are made of proteins and rRNA
- In cytoplasm, there's another type of special RNA that participates in protein synthesis called **tRNA**
- tRNA or Transfer RNA is very different in form and function from mRNA
- Also referred to as sRNA or Soluble RNA

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Translation of Genetic Material

- tRNA has segments which are both double stranded and single stranded
- Double strands are not different as in DNA but areas where different parts of tRNA molecule fold over and base pair
- Result is a folded structure that resembles a clover leaf

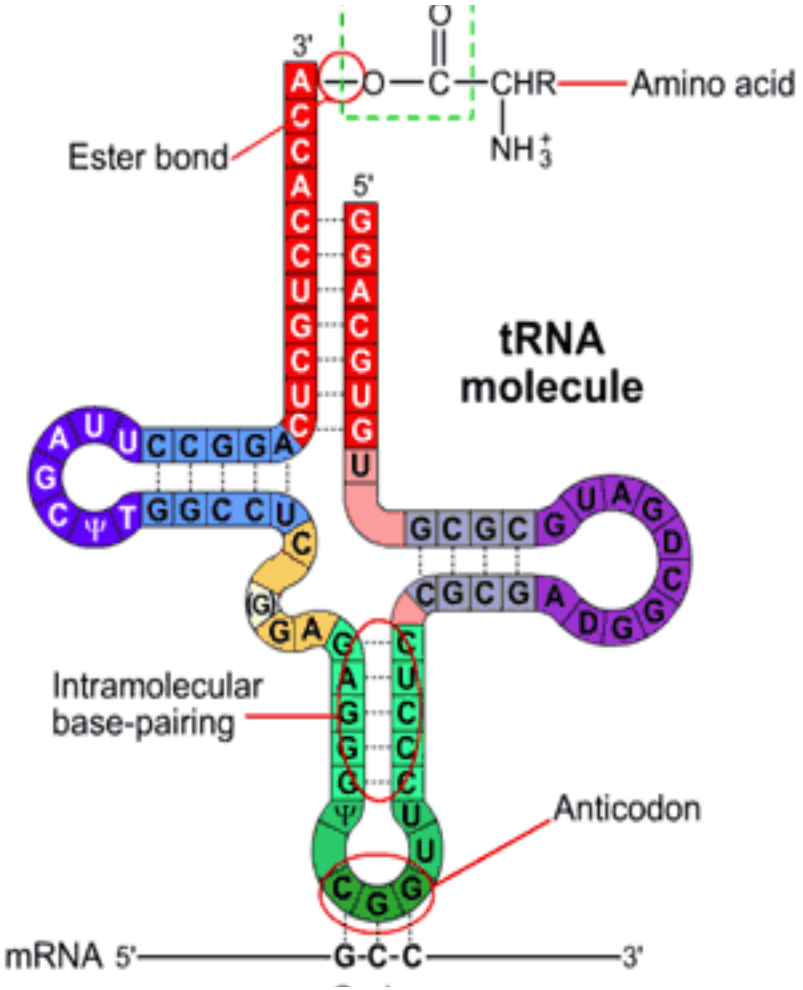


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Translation of Genetic Material

- One end of the tRNA has the sequence CCA
- It is joined at the CCA end by a single molecule of an amino acid
- At the apex of the one of the loops is a sequence of bases called **anticodon** which is the area of attraction between tRNA & mRNA
- The anticodon is complementary to the sequence of bases on mRNA
- Other two loops are place of tRNA's attachment to ribosomes

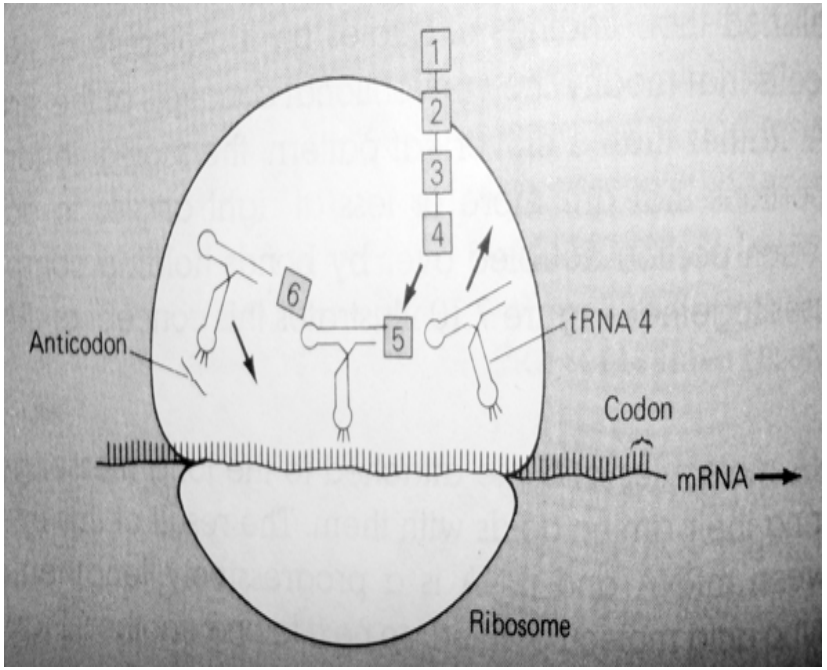


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Translation

- mRNA gets attached to several ribosomes called **polysomes or polyribosomes**
- Several tRNA molecules become attached to long mRNA, and they bring their amino acids with them.
- The result of this interaction between mRNA and tRNA is progressively lengthening series of amino acid molecules lined up next to one another through peptide bonds
- This interaction is a gene-determined protein synthesis known as **translation.**



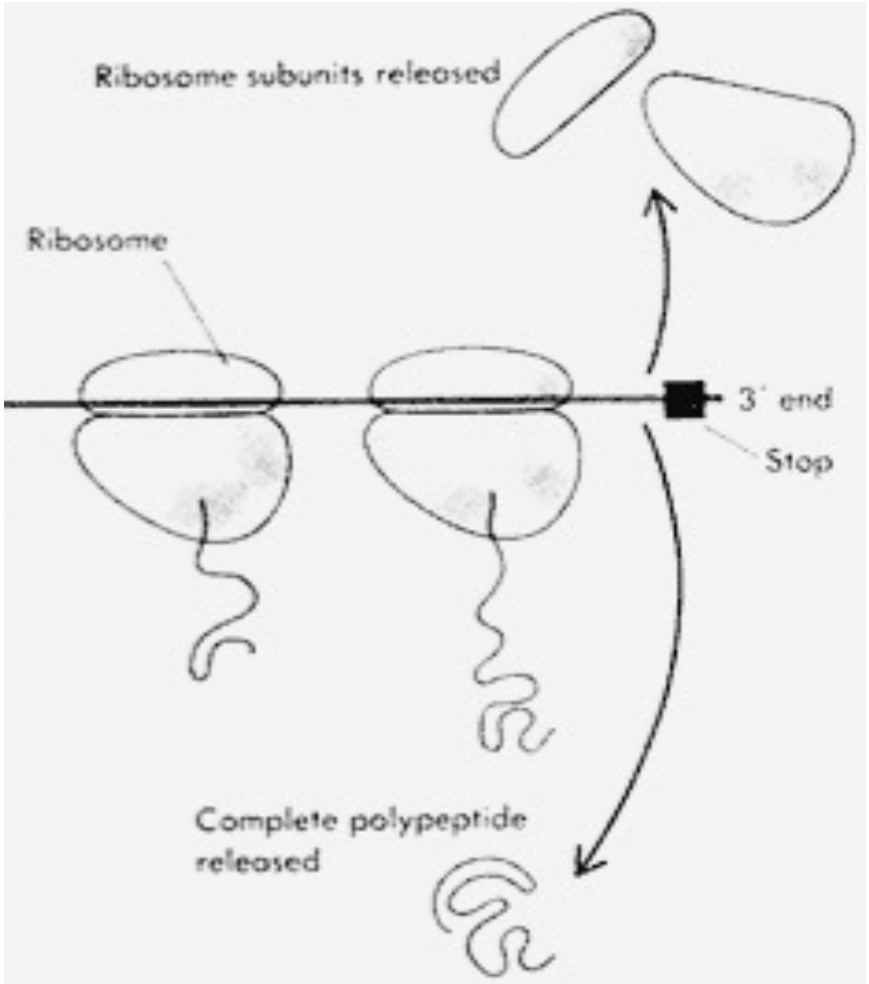
The sequence of amino acids is determined by triplet codon on mRNA

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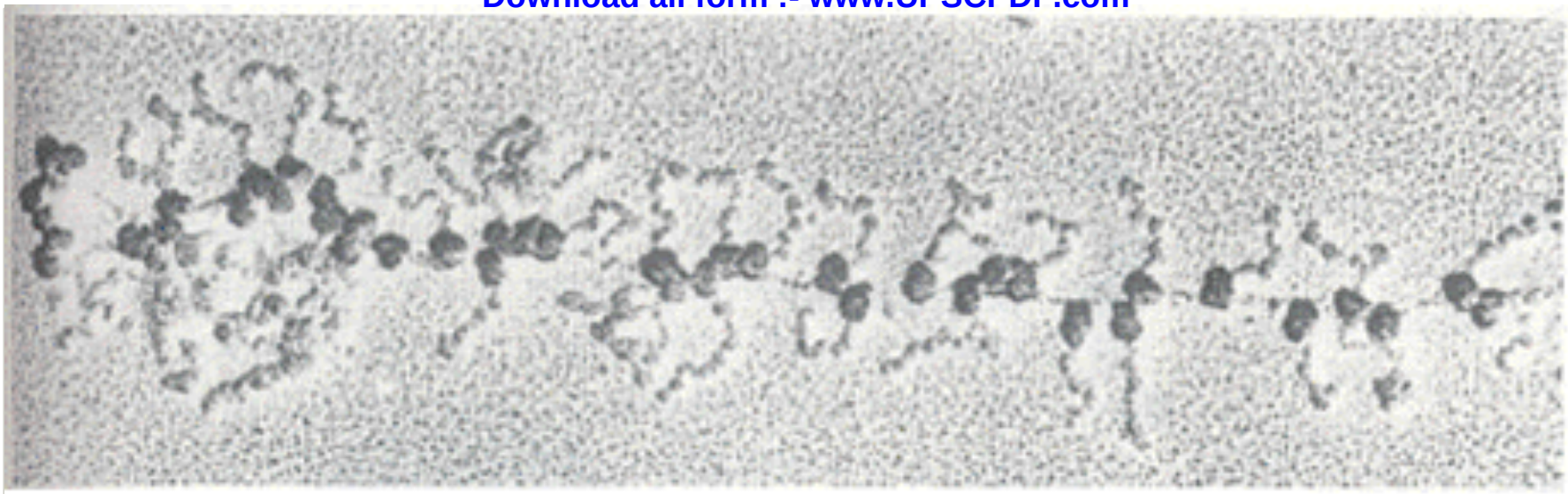
Translation

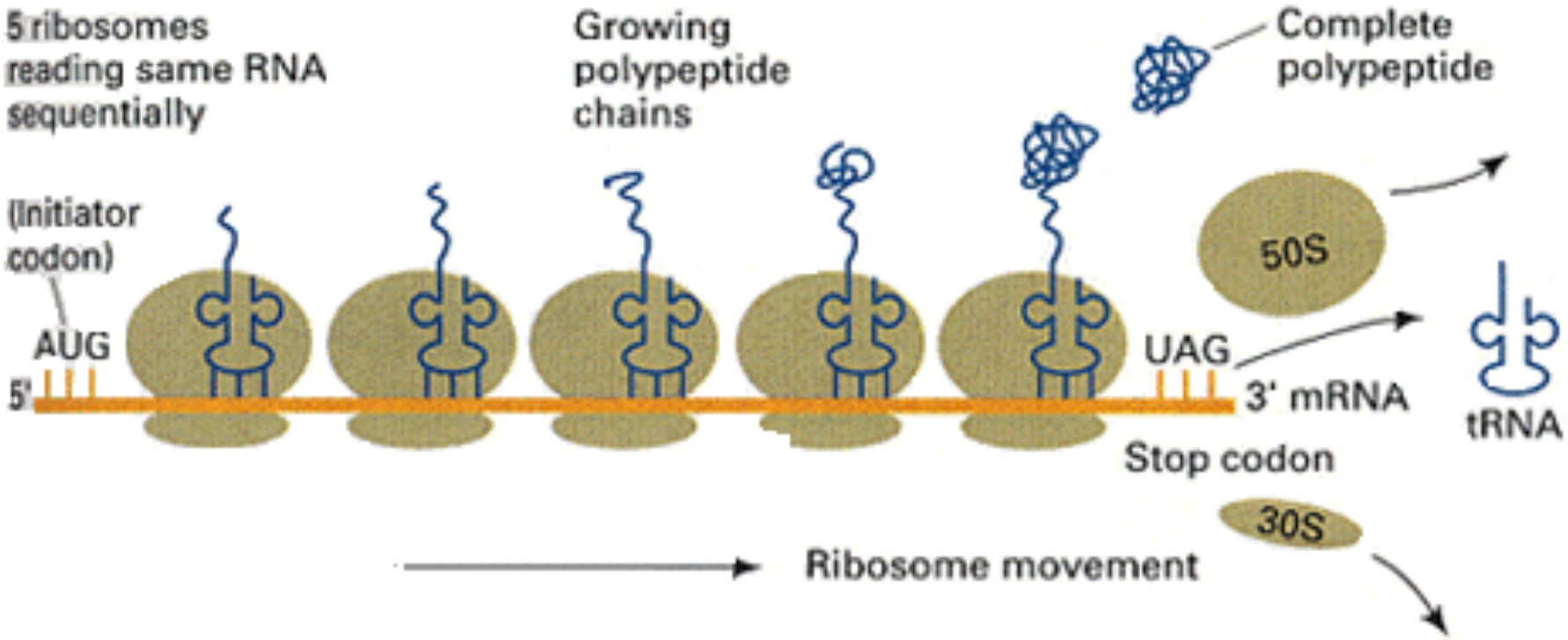
- Once peptide bonds are established between amino acids, they are released by tRNA forming a chain of amino acids called **polypeptide**
- Once the amino acid is released, the tRNA returns to the cytoplasm and fetch another amino acid of the same kind
- Once message is fully translated, the polypeptide chain is released
- Ribosomes break into two parts, which reassemble when polysomes are formed for new protein synthesis



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Other aspects of P/S

- Most mRNAs are short-lived – destroyed once their messages are translated
- It is necessary to waste them as their presence may result in continuous translation of a particular protein disturbing the balance of the cell and are dangerous – **Ex. Adrenalin !!**#@!!**
- If a protein is continuously needed, then more mRNAs are continuously synthesized – **feedback** from cytoplasm to nucleoplasm
- Some mRNAs for proteins like Hemoglobin are stable as this is stored in abundant quantity in RBCs
- tRNAs and ribosomes are much stable & take part in multiple functions

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The Genetic Code

- The messages for a particular protein must be encoded in nitrogenous bases as the sugar and phosphate are essentially same in all cells and all DNA
- There are 20 Amino Acids; and there are 4 Bases in DNA
- Can the code be a double code?
- Then there will be only 16 combinations like A-T, A-C, A-G, A-A, C-T, C-C, C-G, G-C etc. (remember we are talking about sequence of bases on one strand of DNA)
- Whatever the code is, it has to allow for at least 20 different combinations for 20 different amino acids

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The Genetic Code

- The code has to be a **Triplet Code** as this accounts for 64 combinations
- Scientists discovered the code is a triplet code and have interpreted all of them
- Ex. UUU – Phenylalanine; AGU – Serine etc.
- Each triplet representing an amino acid is a **codon**

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First position (5' end)	U	C	A	G	Third position (3' end)
U	UUU Phe UUC UUA Leu UUG	UCU UCC UCA UCG Ser	UAU Tyr UAC UAA Stop UAG Stop	UGU Cys UGC UGA Stop UGG Trp	U C A G
C	CUU CUC CUA CUG Leu	CCU CCC CCA CCG Pro	CAU His CAC CAA Gln CAG	CGU CGC CGA CGG Arg	U C A G
A	AUU AUC AUA Ile AUG	ACU ACC ACA ACG Thr	AAU Asn AAC AAA Lys AAG	AGU Ser AGC AGA Arg AGG	U C A G
G	GUU GUC GUA GUG Val	GCU GCC GCA GCG Ala	GAU Asp GAC GAA Glu GAG	GGU GGC GGA GGG Gly	U C A G

Amino acid names:

Ala = alanine
Arg = arginine
Asn = asparagine
Asp = aspartate
Cys = cysteine
Gln = glutamine
Glu = glutamate
Gly = glycine
His = histidine
Ile = Isolevcine
Leu = leucine
Lys = lysine
Met = methionine
Phe = phenylalanine
Pro = proline
Ser = serine
Thr = threonine
Trp = tryptophan
Tyr = Tyrosine
Val = valine

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The Genetic Code

- 64 possible combinations are more than enough for 20 amino acids
- There is a certain redundancy in genetic code as more than one type of triplet codon exists for a amino acid
- It has been found by Francis Crick that only the first two bases are important for codon and the third base can be any of the four and the codon still translates into the same amino acid
- This is called **Wobble Hypothesis**

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Message Termination

- Chromosomes contain numerous genes and each gene can run into scores of nucleotides
- What causes the transcription of mRNA to occur only over the length of a gene and no overlap into the neighboring gene or DNA?
- It is believed that the codons at the end of the genes do not code for any amino acids
- These codons are called **nonsense codons**

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